Problem #1

A fault analysis was done on a large power system using the symmetrical component method. At one of the loads in the system, the symmetrical components of the line to neutral voltages were computed and are given below. The load is Y-connected with a solidly grounded neutral and per phase impedance of $10\angle 65°\Omega$.

$$\vec{V}_0 = 9.122643\angle -91.77°\,V, \qquad \vec{V}_1 = 113.043087\angle 5.29°\,V, \qquad \vec{V}_2 = 7.396254\angle -169.78°\,V$$

a. Calculate the actual line to neutral voltages and line to line voltages at this load.
b. Calculate the line currents and neutral current at the load.
c. Calculate the real and reactive power consumed by the load.

Problem #2

A 3ϕ generator in an electric utility power system is Y-connected with a solidly grounded neutral. The generator operates with balanced and symmetric voltage outputs with a 28.75KV line to line magnitude and ABC phase sequence. However the generator output line currents are unbalanced. The symmetrical components of the generator output currents are given below.

$$\vec{I}_0 = 186.4262\angle -149.71°A, \qquad \vec{I}_1 = 874.7977\angle 12.86°A, \qquad \vec{I}_2 = 103.3159\angle 21.77°A$$

a. Calculate the actual line currents and the neutral current at this generator.
b. Calculate the total real and reactive power output of this generator, assuming that the voltage between A-phase and ground is a reference phasor (phase angle of zero degrees).

Problem #3

A 3ϕ Y-connected generator has its neutral point tied to ground through a 150Ω pure resistance. During a fault condition, one of the generator's lines is open circuited while the other two lines are short circuited to ground. The generator line currents during this condition are given below.

$$\vec{I}_A = 0\angle 0°A, \qquad \vec{I}_B = 1000\angle 150°A, \qquad \vec{I}_C = 1000\angle 30°A$$

a. Calculate the symmetrical components of the generator line currents.
b. Calculate the neutral current and the voltage between the generator neutral and ground.

Problem #4

A fault analysis was done on a large power system, and the per unit symmetrical components of the current at one of the power system generators are given below.

$$\vec{I}_0 = 0.001555\angle 80.89°\ puA, \qquad \vec{I}_1 = 0.094596\angle 13.89°\ puA, \qquad \vec{I}_2 = 0.092042\angle -45.88°\ puA$$

a. Calculate the actual line currents and the neutral current at this generator, assuming that the base for the per unit calculations is 100MVA and 20KV at the generator.
   Hint: (Base A) = (1000*(Base MVA))/((sqrt(3)*(base KV))
b. From the calculations in part a, can you tell if this generator Y-connected or Δ-connected? If so then what is the generator's connection scheme and why?
c. Is there a fault in the power system that this generator is supplying? If so then why, and what specific type of fault is it?

Problem #5
A 3ϕ generator in an electric utility power system is Y-connected with a hard grounded neutral. The generator operates with balanced and symmetric voltage outputs with a 28.75KV line to line magnitude. However the generator output line currents are unbalanced. The symmetrical components of the generator output currents are given below.

$$\vec{I}_0 = 152.7525\angle -10.893° A, \qquad \vec{I}_1 = 11,720.637\angle -29.511° A, \qquad \vec{I}_2 = 11,431.973\angle 149.499° A$$

a. Calculate the actual line currents and the neutral current at this generator.
b. Calculate the total real and reactive power output of this generator, assuming that the voltage between A-phase and ground is a reference phasor (phase angle of zero degrees).

Problem #6
Measurements of the actual L-N voltages at a load in a faulted power system were made, and the measured values are given below.

$$\vec{V}_{AN} = 2400\angle 100° V, \qquad \vec{V}_{BN} = 1200\angle -80° V, \qquad \vec{V}_{CN} = 1200\angle -80° V$$

a. Calculate the symmetrical components of these unbalanced load voltages.
b. What type of fault is present in this power system? Why?

Problem #7
A 3ϕ generator in an electric utility power system is Δ-connected and operates with balanced and symmetric voltage outputs with a 27.350KV line to line magnitude. However the generator output line currents are unbalanced. The symmetrical components of the generator output currents are given below.

$$\vec{I}_0 = 0\angle 0° A, \qquad \vec{I}_1 = 1590\angle 10° A, \qquad \vec{I}_2 = 1080\angle -5° A$$

a. Calculate the actual line currents at this generator.
b. Calculate the total real and reactive power output of this generator, assuming that the voltage between A-phase and ground is a reference phasor (phase angle of zero degrees).
c. Is there a fault in the power system that this generator is supplying? If so then why, and what specific type of fault is it?

Problem #8
A fault analysis was done on a large power system, and the per unit symmetrical components of the current at one of the power system generators are given below.

$$\vec{I}_0 = 0.004666\angle 80.890°\ pu\text{A}, \qquad \vec{I}_1 = 0.283789\angle 13.887°\ pu\text{A}, \qquad \vec{I}_2 = 0.276127\angle -45.884°\ pu\text{A}$$

a. Calculate the actual line currents and the neutral current at this generator, assuming that the base for the per unit calculations is 100MVA and 18KV at the generator.
b. Is there a fault in the power system that this generator is supplying? If so then why, and what specific type of fault is it?